Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
PI Name:	Holland, Murray John Ph.D.		
Project Title:	Generation of Fragile and Strongly Correlated Quantum States in Bose-Einstein Condensed Gases		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
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	In this research program we are investigating the phenomena of strongly correlated quantum states in quantum degenerate alkali gases. Strong interactions between the atoms arise from the presence of Feshbach scattering resonances which is the focus of a significant experimental effort at this time. Experiments in quantum degenerate gases have helped to elucidate many phenomena in superconductors and superfluid Helium systems by providing a possibility for direct observation (e.g. vortex studies) in a novel quantum system. We are extending these links to include strongly correlated systems. Examples of phenomena we are studying in this context include the superfluid to Mott insulator transition and the fractional quantum Hall effect. The objective of this research program is to carry out a number of specific investigations to explore the physics of strongly correlated quantum systems in the dilute gas realm. The necessary prerequisite theoretical foundation has already been developed. The renormalized quantum field theory for treating resonance superfluid systems has been a focus topic where strong agreement between theory and experiment has		

Task Description:	been established. The objective here is to extend this general framework to incorporate the areas of quantum Hall physics and quantum phase transitions which are important and of broad interest in condensed matter physics. The specific problems to be addressed are the treatment of resonant scattering in optical lattices (in which atoms move in an optical potential pattern formed by intersecting laser beams), in highly rotating quantum gases, and in strongly coupled fermion superfluids. If successful, this research program will have significant implications for understanding a broad spectrum of physical phenomena. Superfluidity is a subject in physics of fundamental importance, with ties to macroscopic quantum phenomena, to superconductivity, and to optical lasers. The most important work in the field of Bose-Einstein condensation in dilute gases is yet to come, because the recently demonstrated ability to provide quantum control of the interactions between the atoms allows potentially a new area of quantum state engineering in a quantum superfluid. The first topics to pursue are the strongly interacting systems relevant to condensed matter physics. These considerations are of such fundamental nature that it seems likely that technologies will be spawned of industrial importance; in metrology, in remote sensing, in quantum logic, and in mesoscale physics, to name a few.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cold quantum atomic systems have potential impact in a wide range of precision metrology and inteferometric measurement systems with many important applications to both earth and space sciences. In much the same way as the laser has impacted a wide range of technological fields, atomic condensates, which form a coherent matter equivalent to the coherent light, may potentially be useful in a broad range of applied phenomena. This will include both contribution to the fundamental understanding of technologically important systems (such as high Tc superconducting materials) by replicating the physical mechanisms in a highly controllable quantum system, and the ability to control precision measurements through the development of atomic lasers, atomic interferometry, and high-precision spectroscopy, in an unprecedented manner.
Task Progress:	This has been a particularly important year with an explosion of interest in the properties of Feshbach resonances in ultracold gases. The basic idea on which the proposal for this research project was born was the fact that scattering resonances, and in particular Feshbach resonances, may be utilized to control interactions in ultracold coherent gases at the quantum level. In addition, the interactions may be extremely large, leading to breakdown of the conventional theoretical formulation and the emergence of extremely interesting and novel states of matter. These proposals over the past year have begun to be experimentally realized, with a number of groups presenting observations of the formation of molecular condensation and fermionic superfluidity. For the first time, a system has been demonstrated in which one can dynamically tune from fermionic superfluid behavior to bosonic superfluid behavior. We have completed our first projects on these ultracold gases leading to four publications in reviewed journals. Invited talks in 2004 have been given at the Workshop on Ultracold Fermi Gases in Levico, Italy, the International Quantum Fluids and Solids meeting in Trento, Italy, the Gordon Research Conference on Correlated Electron Systems in Massachusetts, the International Conference on Atomic Physics in Rio de Janeiro, and the Quantum Enabled Science and Technology workshop in Santa Fe. In addition, we have laid the groundwork for making the connection of bosonic and fermionic superfluidity, and studying the effects of Feshbach resonances in optical lattices and highly rotating gases. This has involved the analytic development of the effective field theory, which is highly nontrivial in the strongly interacting regime and is now complete, and the first numerical studies of the solutions. Ongoing work now will focus on comparison with the experimental data which is just beginning to emerge.
Bibliography Type:	Description: (Last Updated: 02/19/2008)
Articles in Peer-reviewed Journals	Holland, M. "Atomic Beads on Strings of Light" Nature, Vol 429, pp 251-253 (2004) , May-2004
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